

CSCI 3052U: Machine Learning I — Group Project Proposal (Milestone 1, updated for Milestone 2)

Group 3 — Kharintirasakar Uthayanan, David Zajac, Diana Riyahi, Shawn Xiao, Gamze Esen-Erdemir

Changes since Milestone 1

- Language-detection data: the CantoNLU release includes official train/validation/test splits (42,753 / 2,425 / 2,341 sentences), so no split has to be constructed. The splits are grouped by source sentence, which we verified keeps parallel versions of the same sentence in the same split. The main source of the data was the CantoNLU repo at (github.com/Aatlantise/canto-nlu).
- Success criterion and general audience: For each task, success will be reported if the model exceeds the naive benchmarks, this model is intended for practitioners working on low resource languages with limited compute who are looking for a cheap classical model as well as Lee Lab and CantoNLU researchers who want a simple pretrained-model.
- Task definition: language detection is a three-class task (Cantonese, Mandarin, code-mixed), as defined in the CantoNLU paper. We keep all three classes so that our results are comparable with the published ones; a two-class version may be run as an ablation.
- License: the CantoNLU authors confirmed by email that the data and code are released under CC BY 4.0. All derivative work will be attributed to them.
- Risks and limitations: CantoNLU has inconsistent label, synthetic text and differences between the released data and published results which may have an effect on the models performance making cap score comparisons less reliable. Test-set leakage and time constraints were handled through a locked test set and early deadlines.
- Metric: macro-F1 remains the primary metric; its justification now also covers the class imbalance of the language-detection data. Because the paper does not state which F1 average it reports, micro- and weighted-F1 will also be reported when comparing with published results.
- Feasibility and Compute Plan: Required hardware details as well as the information on which this project was tested on was included. There is no GPU required to run this.
- Initial EDA for both tasks is complete (notebooks/ in the repository), and the data-quality findings are documented in DATA_CARD.md. The success criterion (Section 4) now includes a measurable target.

1. Title

Evaluating Classical Machine Learning Baselines on Cantonese Natural Language Understanding

2. Abstract

Cantonese remains relatively under-resourced in natural language processing compared with higher-resource languages. The CantoNLU benchmark was created to evaluate Cantonese natural language understanding across seven tasks and reports results for several pretrained language models. This project will investigate how well simpler machine-learning approaches perform on selected CantoNLU tasks. We plan to focus initially on language detection and sentiment analysis, with a third task considered only if time and resources permit. We will compare naive benchmarks such as majority-class and stratified-random prediction with classical models including Multinomial Naive Bayes and logistic regression using character-level TF-IDF features. We will additionally evaluate one embedding-based comparison model — k-nearest neighbours on fixed multilingual sentence embeddings — which requires inference only, no fine-tuning. Models will be evaluated under a consistent train/validation/test procedure, with macro-F1 as the proposed primary metric (justified by cross-task consistency and per-class visibility; the sentiment set is deliberately balanced, while the language-detection set is imbalanced) and accuracy and per-class measures as secondary metrics. The goal is not necessarily to outperform pretrained language models, but to establish clear, reproducible reference baselines and determine how much performance is gained from more complex pretrained representations. The final project will include a reproducible pipeline, fair model comparison, error analysis, robustness or sensitivity analysis, and a discussion of limitations. If the core comparison completes on schedule, a small pilot evaluation of one generative LLM on one task is retained as a stretch goal aligned with the Lee Lab's stated research direction.

3. Research Background

Natural language understanding (NLU) evaluates whether models can correctly interpret linguistic information in text. The Lee Language Lab's CantoNLU project provides a benchmark containing seven Cantonese NLU tasks and reports

results from existing and newly trained language models. The broader CantoNLU research project is interested in extending evaluation to additional large language models and human performance.

For CSCI 3052U, we propose a narrower and more manageable question: how well can classical machine-learning models perform on selected CantoNLU tasks compared with existing pretrained-model results? Simple baselines provide an important reference point. If a classical model performs close to a pretrained model, this may suggest that useful task-specific patterns can be captured without large-scale pretraining. If the gap is large, it helps quantify the value of pretrained representations. This makes the project useful regardless of which model family performs best.

The main gap we will investigate is the absence of a simple naive-and-classical reference layer in our CantoNLU comparison. Earlier Cantonese sentiment-classification work also motivates the use of character-level features, while modern multilingual models provide a natural higher-complexity reference point. Our project therefore contributes the first missing reference layer of the lab's benchmarking agenda — reproducible naive and classical baselines — while explicitly leaving human-performance measurement to the lab's parallel annotation effort. A small pilot LLM evaluation (one model, one task) is retained as a stretch goal toward the lab's first stated direction, keeping the project substantively aligned with the approved Lee Lab problem.

Research Questions

- RQ1. How close do naive and classical feature-based models come to published CantoNLU pretrained-model results on selected tasks?
- RQ2. Does a fixed multilingual embedding representation improve performance over character-level TF-IDF without requiring full model fine-tuning?
- RQ3. How does model performance change with training-set size, and what types of examples are most often misclassified?

4. Methodology

Data

We use the released CantoNLU task datasets, focusing on language detection and sentiment analysis. A third task may be included only if the first two are completed successfully. Dataset versions, licenses, class distributions, units of analysis, and the official train/validation/test splits were verified during Milestone 2. Sentiment analysis uses OpenRice restaurant reviews with official splits of 9,999 / 999 / 999 reviews and three deliberately balanced classes (smile, ok, cry). Language detection uses official splits of 42,753 / 2,425 / 2,341 sentences and three classes as defined in the CantoNLU paper: Cantonese, Mandarin (converted to traditional characters), and code-mixed sentences generated by randomly replacing tokens with their counterparts from the other language. The code-mixed class makes up about two thirds of every split, and the splits are grouped by source sentence, so no source sentence appears in more than one split. The CantoNLU authors confirmed by email that the data and code are released under CC BY 4.0; this is recorded in the data card. The initial EDA found a small number of language-detection sentences with conflicting labels and a few sentences repeated across splits; we use the released data unchanged so that results remain comparable with the published ones, and report these issues as limitations.

Preprocessing and Models

Text will initially be represented with character-level TF-IDF features, which avoids requiring a separate Cantonese word-segmentation pipeline. We will inspect the data for duplicates, missing or invalid labels, class imbalance, and possible leakage. Our naive benchmarks will include majority-class and stratified-random prediction. Our primary course-level models will be Multinomial Naive Bayes and L2-regularized logistic regression. We will also evaluate k-nearest neighbours using fixed multilingual sentence embeddings (e.g., LaBSE); this requires inference only and runs on a laptop, so it is a committed comparison model rather than a conditional one. More complex models will remain optional until dataset size, compute needs, and feasibility are confirmed.

Experimental Design and Evaluation

We use the official CantoNLU splits for both tasks. Training data will be used to fit models, validation data will be used for model selection and hyperparameter tuning, and the final test set will be reserved for final evaluation. Hyperparameter-search effort will be documented and kept reasonably comparable across models. Published CantoNLU results will be quoted only where our splits and metrics match the benchmark's protocol; otherwise they will be reported as indicative context rather than direct comparisons. Because the CantoNLU paper reports F1 without stating the averaging method, we will report macro-, micro- and weighted-F1 side by side in that comparison.

The proposed primary metric is macro-F1, chosen for cross-task consistency and per-class visibility (the sentiment task is class-balanced, so accuracy is reported alongside it; in the language-detection task the code-mixed class makes up about two thirds of the data, so accuracy alone would be misleading), with accuracy, per-class precision and recall, and confusion matrices as secondary evidence. We will report variability across repeated runs where results are stochastic and will include at least one sensitivity, robustness, or ablation analysis. A likely sensitivity experiment is a learning curve using progressively larger portions of the training data. Error analysis will examine representative misclassified examples and discuss limitations such as domain coverage, dialect coverage, and any dependence on translation or external linguistic tooling.

Success Criterion

The project will be considered successful if we produce a reproducible data and training pipeline, naive benchmark results, at least two substantive machine-learning model results, a fair comparison on at least two CantoNLU tasks, uncertainty or variability evidence where appropriate, and a clear error-and-limitations analysis. Measurably, for each task the best classical model must exceed the naive benchmarks' validation macro-F1, and its gap to the published CantoNLU results will be reported. Success does not depend on outperforming pretrained models. The pilot LLM evaluation is a stretch goal only and does not affect this success criterion.

5. Significance

This project can provide useful reference points for understanding how much performance on Cantonese NLU tasks can be achieved using relatively simple and computationally inexpensive methods. If classical models are competitive on some tasks, they may offer practical low-compute alternatives; if they fall substantially behind, the size of that gap helps quantify the value of pretrained language representations. The work is especially relevant to low-resource-language NLP, where data and compute constraints can make simpler methods valuable. It also fits the goals of Machine Learning I by requiring data preparation, baseline construction, model training, validation, evaluation, reproducibility, error analysis, and evidence-based conclusions.

6. SMART Timeline

Milestone	Specific, measurable target	Due
M1	Finalize proposal, team contract, roles, repository, and Lee Lab topic declaration.	Sept. 18
M2	Verify datasets/licenses; complete initial EDA; confirm tasks, splits, primary metric, risks, and compute plan. Done: official splits and CC BY 4.0 license confirmed.	Oct. 2
M3	Build end-to-end pipeline; implement naive benchmark and first classical baselines; record seeds/environment and baseline table.	Oct. 19
M4	Present preliminary results and theory; identify blockers/deviations; revise timeline and respond to feedback. If on schedule: pilot LLM run (one model, one task; stretch).	Oct. 30
M5	Complete fair model comparison, tuning, variability analysis, error analysis, and one robustness/sensitivity/ablation study.	Nov. 20
M6	Complete 6-8 page report, reproducible repository, main-result regeneration, presentation/demo, contribution and AI-use statements.	Dec. 4
M7	Complete individual reflection, contribution evidence, and oral/code walkthrough.	Dec. 7

7. References

- Min, J., Ng, Y. H., Chan, S., Zhao, H. S., & Lee, E.-S. A. (2025). CantoNLU: A benchmark for Cantonese natural language understanding. arXiv:2510.20670.
- Wang, A., Singh, A., Michael, J., Hill, F., Levy, O., & Bowman, S. R. (2018). GLUE: A multi-task benchmark and analysis platform for natural language understanding. Proceedings of BlackboxNLP 2018.

- Zhang, Z., Ye, Q., Zhang, Z., & Li, Y. (2011). Sentiment classification of internet restaurant reviews written in Cantonese. *Expert Systems with Applications*, 38(6), 7674-7682.
- Xiang, R., Chersoni, E., Li, Y., Li, J., Huang, C.-R., Pan, Y., & Li, Y. (2025). Cantonese natural language processing in the transformers era: A survey and current challenges. *Language Resources and Evaluation*, 59(2), 1747-1773.
- Lee, J. L., Chen, L., Lam, C., Lau, C. M., & Tsui, T.-H. (2022). PyCantonese: Cantonese linguistics and NLP in Python. *Proceedings of LREC 2022*.
- Feng, F., Yang, Y., Cer, D., Arivazhagan, N., & Wang, W. (2022). Language-agnostic BERT sentence embedding. *Proceedings of ACL 2022*, 878–891.